

Course One

Foundations of Data Science



Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- Complete the questions in the Course 1 PACE strategy document
- Answer the questions in the Jupyter notebook project file
- Complete coding prep work on project's Jupyter notebook
- Summarize the column Dtypes
- Communicate important findings in the form of an executive summary

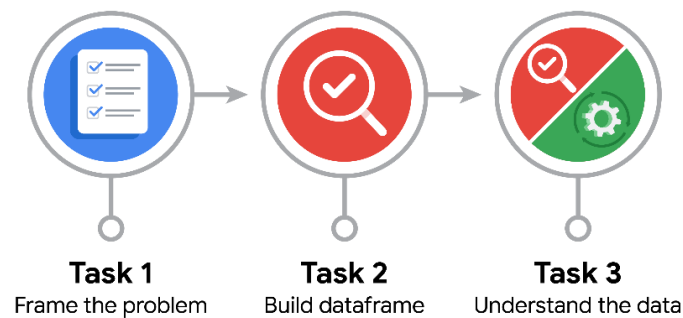
Relevant Interview Questions

Completing the end-of-course project will help you respond these types of questions that are often asked during the interview process:

- Describe the steps you would take to clean and transform an unstructured data set.
- What specific things might you look for as part of your cleaning process?
- What are some of the outliers, anomalies, or unusual things you might look for in the data cleaning process that might impact analyses or ability to create insights?

Reference Guide

This project has three tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations



PACE: Plan Stage

- How can you best prepare to understand and organize the provided information?

To better prepare to understand and organize the data, I first explore the data dictionary, display a few rows using the Pandas function “data.head(5)”, and check for nulls, duplication, and outliers.

- What follow-along and self-review codebooks will help you perform this work?

To perform this analysis, I used Jupyter Notebook, as I have the ability to run code and explain its outputs. I imported pandas and numpy for operations, data manipulation, and insight extraction.

- What are some additional activities a resourceful learner would perform before starting to code?

Resourceful learners should get themselves familiar with the basics of the Python programming language for data analysis. This does not mean that they should know every single function used for data analysis, as that is a lifelong process. For instance, I have never used the “agg” function before and did not really know what arguments should be passed to it. I searched for the documentation, read a bit, and searched for a YouTube tutorial to enhance my understanding. Once I found myself familiar with its usage, I came back to Jupyter and continued my analysis.



**PACE: Analyze Stage**

- Will the available information be sufficient to achieve the goal based on your intuition and the analysis of the variables?

Yes, the available data seems sufficient to achieve the initial goal. However, when building the model, we have to be aware that some variables (.,e., post-publication features) may cause model overfitting, meaning the model would perform very well on the training data, but fall short when new, unseen data is introduced. For these reasons, I focused on features such as author ban status, verified accounts, and video transcription text as the predictors for the target variable, 'claim status'. It is important to note that the model is an iterative asset that should be consistently refined as more data is collected.

- How would you build summary dataframe statistics and assess the min and max range of the data?

For building summary data frame statistics and assessing the min and max, I would use Pandas 'describe()'function.

- Do the averages of any of the data variables look unusual? Can you describe the interval data?

Yes, the averages of the post-publication metrics (view counts, like counts, share counts, and comment counts) are much higher than the median, which indicates the presence of outliers.

**PACE: Construct Stage**

Note: The Construct stage does not apply to this workflow. The PACE framework can be adapted to fit the specific requirements of any project.



PACE: Execute Stage

- Given your current knowledge of the data, what would you initially recommend to your manager to investigate further prior to performing exploratory data analysis?

I would recommend further investigation of the extreme values and the 298 null values to see if they are missing at random or if they follow a pattern, such as being tied to a specific account type.

- What data initially presents as containing anomalies?

Based on the data provided, the engagement metrics, such as view counts, like counts, and share counts, were presented with extreme outliers that made the data highly right-skewed.

- What additional types of data could strengthen this dataset?

More data about the authors, such as the account age and the number of followers, as well as the audio features, such as the emotional tone, background music, and the objects displayed in the video, could strengthen this dataset even further.